DEVOPS - 04-03-2026

Prune cmd used to remove all file that are exited or half build

```
C:\Users\HP>docker system prune
WARNING! This will remove:
  - all stopped containers
  - all networks not used by at least one container
  - all dangling images
  - unused build cache
```

Give y then only it will delete the files

```
Are you sure you want to continue? [y/N] n

C:\Users\HP>docker system prune
WARNING! This will remove:
  - all stopped containers
  - all networks not used by at least one containe
  - all dangling images
  - unused build cache
```

```
Deleted build cache objects:
nnp1wwrpzmnbo7uwuta15756e
wv1t9f0k5az2y9axnwgdcluij
4uenwh5f2ec7rr71bq1hy5qkg
6b0yos59o4f3gn5jdgem7wex3
skethq74vnzgb19q8cc7f47ng
xyt1cy4nb74xx66z4s3ixq7dd

Total reclaimed space: 17.9MB
```

Everything where deleted from docker

```
C:\Users\HP>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

ls used to list the network

```
C:\Users\HP>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
f073c4da6582   bridge    bridge    local
b7a2fd77fa43   host      host      local
3d303eb80444   none      null      local
```

Creating a network

```
C:\Users\HP>docker network create siet-devops
7e11c08b829efa5e1c241ef1ab0c9d5dc7cb447619586947a45a6bd81958f728
```

ls used to list the network

```
C:\Users\HP>docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
f073c4da6582   bridge        bridge    local
b7a2fd77fa43   host          host      local
3d303eb80444   none          null      local
7e11c08b829e   siet-devops   bridge    local
```

Checking images to create containers

```
C:\Users\HP>docker images
                                                  i Info →   U  In Use
IMAGE                        ID            DISK USAGE   CONTENT SIZE   EXTRA
admin/customwebserver:1.0    2fa91ad76a61       279MB         76.6MB
httpd:latest                 96b1e8f69ee3       177MB         47.6MB
my-img:latest                ff9cbfaaebd2       424MB          118MB
sietbk:1.0                   2fa91ad76a61       279MB         76.6MB
ubuntu:latest                d1e2e92c075e       119MB         31.7MB
```

Creating a container

```
C:\Users\HP>docker run -d -it --name siet --network siet-devops -p 82:80 my-
img:latest
a4836a0a1f4236055c2d1e6c7c051714d291226357de3b41102d70cfed99470a
```

ls used to list the network

```
C:\Users\HP>docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
f073c4da6582   bridge        bridge    local
b7a2fd77fa43   host          host      local
3d303eb80444   none          null      local
7e11c08b829e   siet-devops   bridge    local
```

Creating null, host & bridge network containers

```
C:\Users\HP>docker network create --driver null newnone
Error response from daemon: only one instance of "null" network is allowed

C:\Users\HP>docker network create --driver host newnone
Error response from daemon: only one instance of "host" network is allowed

C:\Users\HP>docker network create --driver bridge newnone
df386fbaaeb80d416845529a7c136f7b51db2fc01693695a46fb885165f88058
```

ls used to list the network

```
C:\Users\HP>docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
f073c4da6582   bridge        bridge    local
b7a2fd77fa43   host          host      local
df386fbaaeb8   newnone       bridge    local
3d303eb80444   none          null      local
7e11c08b829e   siet-devops   bridge    local
```

Docker network rm (networkid networkid) => id vachu remove pandrom

```
C:\Users\HP>docker network rm df386fbaaeb8 7e11c08b829e
df386fbaaeb8
Error response from daemon: error while removing network: network siet-devop
s has active endpoints (name:"siet" id:"026c845fb3ed")
exit status 1

C:\Users\HP>docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
f073c4da6582   bridge        bridge    local
b7a2fd77fa43   host          host      local
3d303eb80444   none          null      local
7e11c08b829e   siet-devops   bridge    local
```

Docker run -d -it –name nonecon –network none -p 82:80 my-img:latest

```
C:\Users\HP>docker run -d -it --name hostcon --network host 82:80 my-img:lat
est
Unable to find image '82:80' locally
docker: Error response from daemon: pull access denied for 82, repository do
es not exist or may require 'docker login'

Run 'docker run --help' for more information
```

Docker run -d -it –name hostcon –network host -p 82:80 my-img:latest

```
C:\Users\HP>docker run -d -it --name hostcon --network host -p 82:80 my-img:
latest
WARNING: Published ports are discarded when using host network mode
435f7840766726bb9b3c3471a7d257e01d13eb5a30250cbff3091dce3b5de2d4
```

Docker exec -it hostcon /bin/bash

Apt update

Exit

```
C:\Users\HP>docker exec -it hostcon /bin/bash
root@docker-desktop:/# apt update
Err:1 http://archive.ubuntu.com/ubuntu noble InRelease
  403  Forbidden [IP: 172.16.3.1 8090]
Err:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
  403  Forbidden [IP: 172.16.3.1 8090]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Err:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
  403  Forbidden [IP: 172.16.3.1 8090]
Get:5 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [126
0 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1894 kB
]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3
300 kB]
Reading package lists... Done
N: See apt-secure(8) manpage for repository creation and user configuration details
.
N: Updating from such a repository can't be done securely, and is therefore disable
d by default.
E: The repository 'http://archive.ubuntu.com/ubuntu noble InRelease' is no longer s
igned.
E: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble/InRelease  403  For
bidden [IP: 172.16.3.1 8090]
E: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble-updates/InRelease
403  Forbidden [IP: 172.16.3.1 8090]
E: The repository 'http://archive.ubuntu.com/ubuntu noble-updates InRelease' is no
longer signed.
N: Updating from such a repository can't be done securely, and is therefore disable
d by default.
N: See apt-secure(8) manpage for repository creation and user configuration details
.
E: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/noble-backports/InRelease
  403  Forbidden [IP: 172.16.3.1 8090]
E: The repository 'http://archive.ubuntu.com/ubuntu noble-backports InRelease' is n
o longer signed.
N: Updating from such a repository can't be done securely, and is therefore disable
d by default.
N: See apt-secure(8) manpage for repository creation and user configuration details
.
root@docker-desktop:/# exit
exit
```

Docker ps

```
C:\Users\HP>docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED          STATUS
      PORTS                                       NAMES
435f78407667   my-img:latest   "/bin/sh -c 'apachec…"   3 minutes ago    Up 3 minut
es                                                 hostcon
a4836a0a1f42   my-img:latest   "/bin/sh -c 'apachec…"   27 minutes ago   Up 27 minu
tes   0.0.0.0:82->80/tcp, [::]:82->80/tcp          siet
```

Docker port (container name)

```
C:\Users\HP>docker port hostcon

C:\Users\HP>docker port siet
80/tcp -> 0.0.0.0:82
80/tcp -> [::]:82

C:\Users\HP>
```

Docker run -d -it --name (name) --network (name) -P (img name)

```
C:\Users\HP>docker run -d -it --name portcon --network newnt -P my-img:latest
5d0348e69e6e652e02838884baaa1d743d778f8eb26ee62683ac9f5ff1c7e566
docker: Error response from daemon: failed to set up container networking: network
newnt not found

Run 'docker run --help' for more information
```

Without port tried but not worked

Docker network inspect bridge

```
C:\Users\HP>docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
f073c4da6582   bridge        bridge    local
b7a2fd77fa43   host          host      local
3d303eb80444   none          null      local
7e11c08b829e   siet-devops   bridge    local
```

Now lets inspect this network

```
C:\Users\HP>docker network inspect siet-devops bridge
[
    {
        "Name": "siet-devops",
        "Id": "7e11c08b829efa5e1c241ef1ab0c9d5dc7cb447619586947a45a6bd81958f728",
        "Created": "2026-03-04T04:00:30.850774308Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {
            "a4836a0a1f4236055c2d1e6c7c051714d291226357de3b41102d70cfed99470a": {
                "Name": "siet",
                "EndpointID": "026c845fb3ed59757d5ad7a9eea789a7fc0391188c6bf1547369
eb54a23fbe83",
                "MacAddress": "da:e7:84:c3:6d:83",
                "IPv4Address": "172.18.0.2/16",
                "IPv6Address": ""
            }
        },
        "Status": {
            "IPAM": {
```

In created network container is not empty but default network container will be empty

```
    "Name": "bridge",
    "Id": "f073c4da6582e5ae5c129243fc171e1179b83ff079a96f7303bd78bce73d70d4",
    "Created": "2026-03-04T03:52:15.400682121Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
        "Driver": "default",
        "Options": null,
        "Config": [
            {
                "Subnet": "172.17.0.0/16",
                "Gateway": "172.17.0.1"
            }
        ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
        "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
        "com.docker.network.bridge.default_bridge": "true",
        "com.docker.network.bridge.enable_icc": "true",
        "com.docker.network.bridge.enable_ip_masquerade": "true",
        "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
        "com.docker.network.bridge.name": "docker0",
        "com.docker.network.driver.mtu": "1500"
    },
    "Labels": {},
    "Containers": {},
    "Status": {
        "IPAM": {
            "Subnets": {
                "172.17.0.0/16": {
                    "IPsInUse": 3,
                    "DynamicIPsAvailable": 65533
```

General volume

ls used to list the volume

```
C:\Users\HP>docker volume ls
DRIVER    VOLUME NAME
```

Create a volume

```
C:\Users\HP>docker volume create siet-testvol
siet-testvol

C:\Users\HP>docker volume ls
DRIVER    VOLUME NAME
local     siet-testvol
```

Mount a volume while creating a container

```
C:\Users\HP>docker run -d -it --name voltest --mount source=siet-testvol,target=/data httpd
ee621caede8c94cbe7ff0756c6a46201089072c875ae70db88de9f7d85b04284
```

Executing the container in which volume is mount

```
C:\Users\HP>docker exec -it voltest /bin/bash
root@ee621caede8c:/usr/local/apache2# cd /data
root@ee621caede8c:/data#
```

Inspect it

```
C:\Users\HP>docker inspect voltest
[
    {
        "Id": "ee621caede8c94cbe7ff0756c6a46201089072c875ae70db88de9f7d85b04284",
        "Created": "2026-03-04T05:55:16.863800419Z",
        "Path": "httpd-foreground",
        "Args": [],
        "State": {
            "Status": "running",
            "Running": true,
```

The mount volume will shown

```
  "Mounts": [
      {
          "Type": "volume",
          "Name": "siet-testvol",
          "Source": "/var/lib/docker/volumes/siet-testvol/_data",
          "Destination": "/data",
          "Driver": "local",
          "Mode": "z",
          "RW": true,
          "Propagation": ""
      }
  ],
```

Make a directory

```
C:\Users\HP>mkdir siet_bindmount
```

It is shown in dir

```
11-10-2025  12:21    <DIR>          Searches
02-03-2026  14:25        31,686,656 sietbackup.tar
04-03-2026  11:36    <DIR>          siet_bindmount
24-12-2025  14:58    <DIR>          Videos
06-11-2025  18:58    <DIR>          WPS Cloud Files
```

Open the directory created

```
C:\Users\HP>cd siet_bindmount
```

Create a container bound data from old container

```
C:\Users\HP\siet_bindmount>docker run -d -it --name voltest2 -v C:\Users\HP\siet_bindmount:/data my-img:latest
e269abc174f3840540f93ed95fe6c2399b9e4874f8fd6a31b39679c728ee8211
```

Executing the container

```
C:\Users\HP\siet_bindmount>docker exec -it voltest2 /bin/bash
root@e269abc174f3:/# cd /data
root@e269abc174f3:/data# apt update -y
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [1
Err:2 http://archive.ubuntu.com/ubuntu noble InRelease
  403  Forbidden [IP: 172.16.2.1 8000]
```

apt install vim -y

After updating and installing vim create a file and exit

```
root@e269abc174f3:/data# vi test.html
```

```
root@e269abc174f3:/data# exit
exit
```

Now in dir the created file will be shown

inspect used to show the detailed view of volume

```
C:\Users\HP\siet_bindmount>docker volume inspect siet-testvol
[
    {
        "CreatedAt": "2026-03-04T05:53:35Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/siet-testvol/_data",
        "Name": "siet-testvol",
        "Options": null,
        "Scope": "local"
    }
]
```

--help used to know the available commands in that

```
\Users\HP\siet_bindmount>docker volume --help
age:  docker volume COMMAND

nage volumes

mmands:
create      Create a volume
inspect     Display detailed information on one or more volumes
ls          List volumes
prune       Remove unused local volumes
rm          Remove one or more volumes

n 'docker volume COMMAND --help' for more information on a command.
```

```
C:\Users\HP\siet_bindmount>docker network --help
Usage:  docker network COMMAND

Manage networks

Commands:
  connect     Connect a container to a network
  create      Create a network
  disconnect  Disconnect a container from a network
  inspect     Display detailed information on one or more networks
  ls          List networks
  prune       Remove all unused networks
  rm          Remove one or more networks

Run 'docker network COMMAND --help' for more information on a command.
```

Docker compose

Reference file:
https://docs.docker.com/reference/compose-file

```
C:\Users\HP>docker-compose --version
Docker Compose version v5.0.2
```

Reference => version and name attributes

In cmd
Make a directory and move to that directory

```
C:\Users\HP>mkdir compose

C:\Users\HP>cd compose
```

Creating a yaml file in notepad

```
C:\Users\HP\compose>notepad docker-compose.yaml
```

Write this code inside yaml file

docker-compose.yaml

File Edit View

```yaml
services:
  web:
    image: httpd:latest
    container_name: my-httpd
    ports:
      - "81:80"
    volumes:
      - ./html:/usr/local/apache2/htdocs/

  db:
    image: mysql:latest
    container_name: my-mysql
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: mydatabase
      MYSQL_USER: user
      MYSQL_PASSWORD: password
    ports:
      - "3306:3306"
    volumes:
      - mysql_data:/var/lib/mysql

volumes:
  mysql_data:
```

Go to directory and check the file is there

```
C:\Users\HP\compose>dir
 Volume in drive C has no label.
 Volume Serial Number is 04E3-B944

 Directory of C:\Users\HP\compose

04-03-2026  14:18    <DIR>          .
04-03-2026  14:17    <DIR>          ..
04-03-2026  14:18                 0 docker-compose.yaml
               1 File(s)              0 bytes
               2 Dir(s)  410,996,895,744 bytes free
```

docker -compose up => used to compose the file

```
C:\Users\HP\compose>docker-compose up
time="2026-03-04T14:43:59+05:30" level=warning msg="C:\\Users\\HP\\compose\\docker-compose.
yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid po
tential confusion"
[+] up 17/17
 ✔Image mysql:latest        Pulled                          672.3s
 ✔Network compose_default   Created                         0.1s
 ✔Volume compose_mysql_data Created                         0.0s
 ✔Container my-httpd        Created                         0.3s
 ✔Container my-mysql        Created                         0.3s
Attaching to my-httpd, my-mysql
my-mysql  | 2026-03-04 09:25:13+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Serv
er 9.6.0-1.el9 started.
my-httpd  | AH00558: httpd: Could not reliably determine the server's fully qualified domai
n name, using 172.19.0.2. Set the 'ServerName' directive globally to suppress this message
```

docker -compose up -d => used to run in background

```
C:\Users\HP\compose>docker-compose up -d
time="2026-03-04T15:01:18+05:30" level=warning msg="C:\\Users\\HP\\compose\\docker-compose.
yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid po
tential confusion"
[+] up 2/2
 ✔Container my-httpd Running                                                    0.0s
 ✔Container my-mysql Running                                                    0.0s

C:\Users\HP\compose>
```

Docker la ps koodutha run agura ellam container yum kattum ithula compose muliyama create pannatum kathum docker-compose shows only compose la create pannunadhu

```
C:\Users\HP\compose>docker-compose ps -a
time="2026-03-04T15:13:13+05:30" level=warning msg="C:\\Users\\HP\\compose\\
docker-compose.yaml: the attribute `version` is obsolete, it will be ignored
, please remove it to avoid potential confusion"
NAME        IMAGE           COMMAND                  SERVICE   CREATED
  STATUS          PORTS
my-httpd    httpd:latest    "httpd-foreground"       web       18 minutes ago
  Up 16 minutes   0.0.0.0:81->80/tcp, [::]:81->80/tcp
my-mysql    mysql:latest    "docker-entrypoint.s…"   db        18 minutes ago
  Up 16 minutes   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp

C:\Users\HP\compose>docker ps -a
CONTAINER ID   IMAGE           COMMAND                  CREATED          STA
TUS          PORTS                                          NAMES
6dda0d255025   httpd:latest    "httpd-foreground"       18 minutes ago   Up
16 minutes   0.0.0.0:81->80/tcp, [::]:81->80/tcp            my-httpd
66f568b9f0af   mysql:latest    "docker-entrypoint.s…"   18 minutes ago   Up
16 minutes   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp    my-mysql
e269abc174f3   my-img:latest   "/bin/sh -c 'apachec…"   4 hours ago      Up
4 hours                                                     voltest2
ee621caede8c   httpd           "httpd-foreground"       4 hours ago      Up
4 hours      80/tcp                                         voltest
5d0348e69e6e   my-img:latest   "/bin/sh -c 'apachec…"   5 hours ago      Cre
ated                                                        portcon
435f78407667   my-img:latest   "/bin/sh -c 'apachec…"   5 hours ago      Up
5 hours                                                     hostcon
a4836a0a1f42   my-img:latest   "/bin/sh -c 'apachec…"   5 hours ago      Up
5 hours      0.0.0.0:82->80/tcp, [::]:82->80/tcp            siet
```

Similarly for images docker images shows all docker-compose images only created will show

```
C:\Users\HP\compose>docker-compose images
time="2026-03-04T15:15:49+05:30" level=warning msg="C:\\Users\\HP\\compose\\
docker-compose.yaml: the attribute `version` is obsolete, it will be ignored
, please remove it to avoid potential confusion"
CONTAINER           REPOSITORY          TAG                 PLATFORM
    IMAGE ID            SIZE                CREATED
my-httpd            httpd               latest              linux/amd64
    96b1e8f69ee3        45.2MB              7 days ago
my-mysql            mysql               latest              linux/amd64
    e5dc14f6e01c        266MB               12 days ago

C:\Users\HP\compose>
```

```
C:\Users\HP\compose>docker-compose -f (filename) up -d
```

```
C:\Users\HP\compose>docker-compose down
time="2026-03-04T15:19:15+05:30" level=warning msg="C:\\Users\\HP\\compose\\
docker-compose.yaml: the attribute `version` is obsolete, it will be ignored
, please remove it to avoid potential confusion"
[+] down 3/3
 ✔Container my-mysql      Removed                                    1.6s
 ✔Container my-httpd      Removed                                    1.5s
 ✔Network compose_default Removed                                    0.3s

C:\Users\HP\compose>
```